



LEGO RESALE INTELLIGENCE

COLLECTOR-GRADE RESALE INTELLIGENCE

LEGO Resale Intelligence

A premium resale intelligence tool for retired LEGO listings.

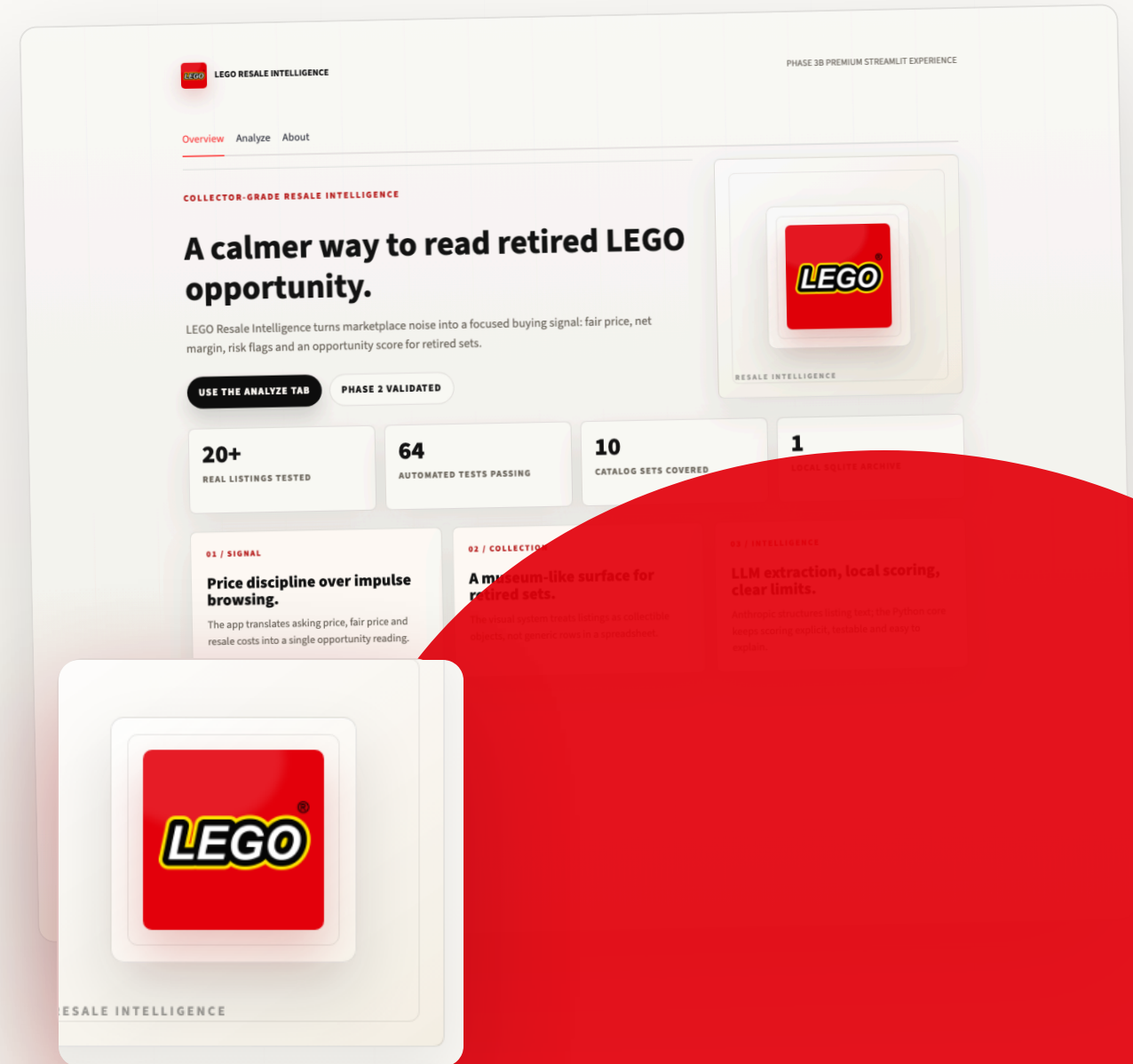
PYTHON

STREAMLIT

ANTHROPIC

SQLITE

PHASE 4 · CASE STUDY



SIGNAL VS NOISE

Marketplace listings are noisy. Good opportunities are easy to miss.

Retired LEGO sets can be valuable, but listings are inconsistent: incomplete titles, unclear condition, missing set IDs, unstable URLs and prices that are hard to compare.

SET ID

Often missing, hidden, translated or mixed with marketplace noise.

PRICE

Asking price is easy to see; real net margin is not.

RISK

Incomplete boxes, unclear condition and missing parts change the deal.

SCORE

A buyer needs a fast way to decide whether to inspect or skip.

PRODUCT FLOW

One URL. One score.

The app turns a real listing URL into a structured buying signal: fair price, net margin, risk flags, opportunity score and local archive.

CAPTURE

EXTRACT

SCORE

ARCHIVE

LEGO

LEGO RESALE INTELLIGENCE

PHASE 3B PREMIUM STREAMLIT EXPERIENCE

Overview

Analyze

About

ANALYZE

LIVE PIPELINE

Paste a listing URL. The system captures the HTML, extracts structured data, identifies the LEGO set, calculates margin and stores the result locally.

LISTING INTAKE

URL ANALYSIS

Paste one listing URL. The system will keep the current flow local, explicit and traceable.

https://es.wallapop.com/item/...

Analyze and save

RECENT INTELLIGENCE

SCORE

SQLITE ARCHIVE

ARCHIVE INSPECTOR

SQLITE DETAIL

Review one saved analysis

#28 · 75192 · GREEN · score 97

ARCHIVED DETAIL

RECORD #28

ARCHIVED SCORE

97

GREEN

SET ID

75192

CONDITION

USED COMPLETE

MARKETPLACE

wallapop

ASKING PRICE

400.00 EUR

FAIR PRICE

720.00 EUR

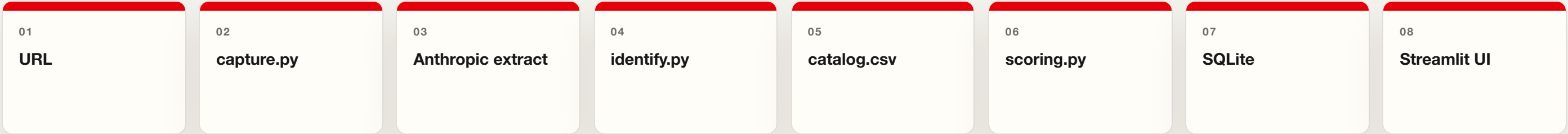
NET MARGIN

240.00 EUR

EXPLAINABLE PIPELINE

A simple system that a junior developer can defend.

No FastAPI, no React, no Postgres, no Docker. The goal was a v1 that can be understood, tested and iterated without enterprise architecture.



NOT MAGIC

Scoring is explicit, explainable and testable.

ARCHIVED DETAIL

RECORD #28

ARCHIVED SCORE 97 GREEN	SET ID 75192	CONDITION USED COMPLETE	MARKETPLACE wallapop
	ASKING PRICE 400.00 EUR	FAIR PRICE 720.00 EUR	NET MARGIN 240.00 EUR

GROSS MARGIN

fair price - acquisition cost

NET MARGIN

minus fees and outbound shipping

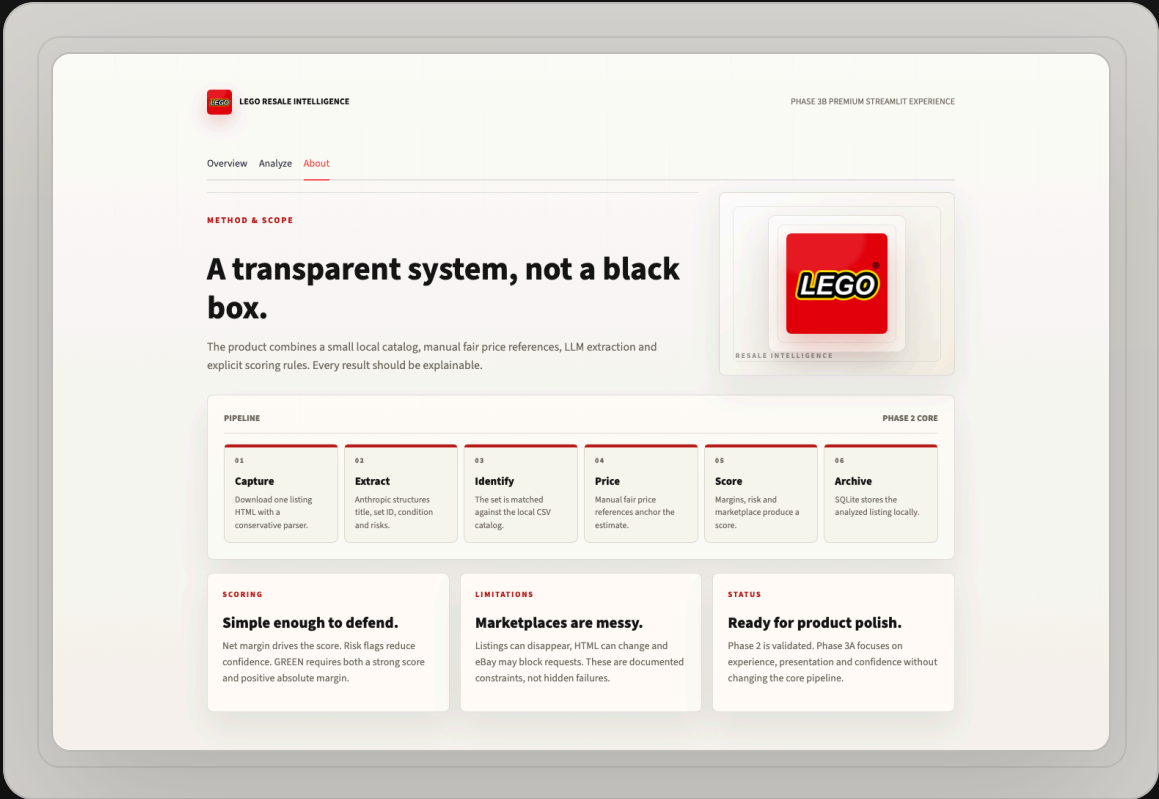
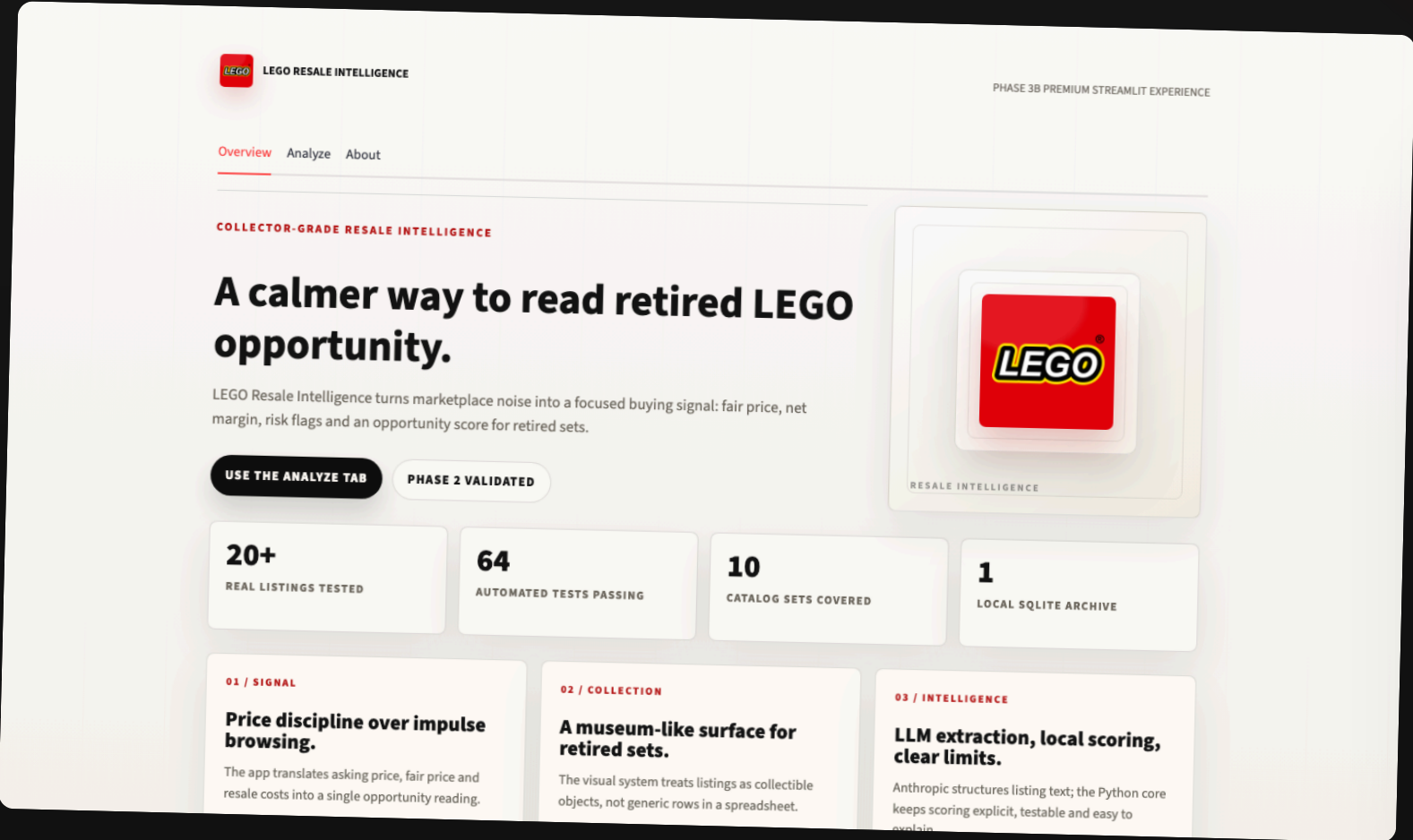
GREEN

strong score + positive margin

VISUAL SYSTEM

Premium UI, grounded in a real workflow.

Phase 3 transformed the prototype into a polished Streamlit experience with Overview, Analyze and About.



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Analyze and save

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MOBILE PROOF

Designed to hold up beyond desktop.

 **LEGO RESALE INTELLIGENCE**

PHASE 3B PREMIUM STREAMLIT EXPERIENCE

Overview

Analyze

About

COLLECTOR-GRADE RESALE INTELLIGENCE

A calmer way to read retired LEGO opportunity.

LEGO Resale Intelligence turns marketplace noise into a focused buying signal: fair price, net margin, risk flags and an opportunity score for retired sets.

USE THE ANALYZE TAB

PHASE 2 VALIDATED

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LISTING INTAKE

URL ANALYSIS

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PHASE 3B PREMIUM STREAMLIT EXPERIENCE

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METHOD & SCOPE

A transparent system, not a black box.

The product combines a small local catalog, manual fair price references, LLM extraction and explicit scoring rules. Every result should be explainable.

EVIDENCE

Tested with real URLs, not only mock data.

Failures were documented as external marketplace constraints, not hidden errors.



PRODUCT ENGINEERING

Built as a serious v1, not an over-engineered system.

CAPTURE

capture.py

LLM

extract.py

CATALOG

identify.py

PRICING

pricing_reference.py

SCORE

scoring.py

FLOW

pipeline.py

STORAGE

db.py

INTERFACE

app/main.py

POLISH

app/styles.py

CLOSING SIGNAL

From noisy marketplace listings to a focused buying signal.

LEGO Resale Intelligence is a personal product engineering project built to show end-to-end thinking: data capture, LLM extraction, structured modeling, scoring, persistence, testing and premium product presentation.

Built by Juan Manuel Muriel Mora
Python · Product Engineering · AI-assisted workflows · UI polish
Personal educational project. Not affiliated with or endorsed by the LEGO Group.

